Phase changes

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for energy transfer during a phase change without a temperature rise.

SUCCESS CRITERIA - I can...

1. Identify a constant-temperature phase-change interval.
2. Use Q=mL for that interval.
3. Separate warming energy from phase-change energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “phase change” and “latent heat” to explain your answer. Sentence starters are optional.

phase change: a change between states such as solid and liquid

latent heat: energy per mass needed for a specified phase change

temperature: a measure of thermal state; it is not an energy amount

01 Warm-up: bring back an earlier idea

Two objects at **280 K** have different masses and materials. Must their internal energies be equal? **Give two reasons**.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

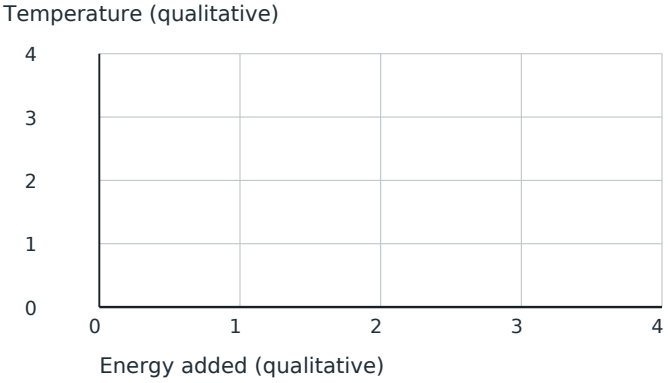
02 Worked example: explain the prediction

Reference: Q = mL; Q = mc ΔT.
Use mL during a phase change at constant temperature, and mc ΔT outside that interval. L is latent heat.

Worked reasoning: Melting **2 kg** of a substance with latent heat **100 J/kg** requires **200 J**. During ideal melting at fixed pressure, added energy changes phase without raising temperature.

Your turn: Sketch temperature versus added energy through warming and melting. **Mark the plateau**.

Sentence starter: In my work, I used … to show … . I labeled it … .

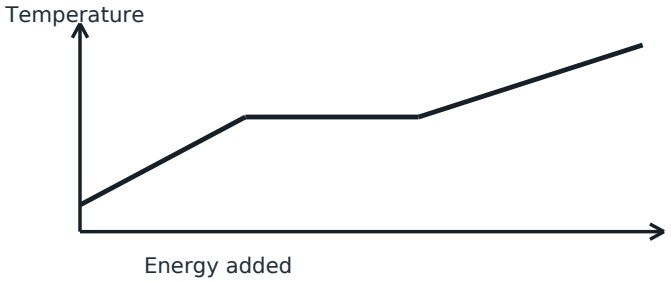


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

On the heating curve, name the flat interval. **Explain why added energy** does not raise temperature during that interval.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



The horizontal axis shows energy added. The vertical axis shows temperature.

Marzano check - circle: ① ② ③

04 Apply the physics

A beaker contains **0.10 kg** of ice at **0°C**. First melt it, then warm the water to **20°C**.
1. Use L_f = **334,000 J/kg** and c_{water} = **4200 J**/(kg·°C).
2. **Label the two stages**, calculate each energy transfer and **find the total**.

Sentence starter: I found … [unit]. This value means … in this situation.

Stage	Equation	Energy (J)
Melt at 0°C	____	____
Warm 0→20°C	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student uses mc ΔT for a melting interval with ΔT=**0** and concludes no energy is required. **Explain the missing term**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

With right positive, a spring has k=**200 N/m** and displacement x=**-0.06 m**. **Find signed spring force** and direction.
Mark equilibrium, displacement and restoring-force direction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

07 Exit check

A **0.20 kg** vapor sample releases **45,000 J** as it condenses to liquid at constant temperature.
1. **Label the outgoing energy arrow**.
2. **Find its latent heat of** condensation and **state the sign of Q** for the sample.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial T / phase: _____ Final T / phase: _____



Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **0.2 kg** ice sample at **0°C** melts and the water warms to **10°C**.
1. Use L_f=**334000 J/kg** and c_{water}=**4200 J**/(kg·°C).
2. **Find total heat**.
3. **Label the energy arrow** and **identify what changes** in the sample.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial T / phase: _____ Final T / phase: _____



Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3